

YASH GOYAL

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SUMMARY

Creative and impact-driven Machine Learning Engineer with a flair for turning cutting-edge tech into deployed, real-world systems. Built GPU-powered ML pipelines integrated with NVIDIA Omniverse at AirLabOne and scaled enterprise analytics in the cloud at SAS. Hackathon winner, university instructor, and multi-disciplinary builder fluent in Python, cloud-native stacks, and full-stack deployment. Whether it's optimizing models, mentoring teams, or shipping production code, I bring a maker's mindset and a mission to solve problems that matter

CORE SKILLS

- **Data Science & ML:** Scikit-Learn, PyTorch, TensorFlow, Keras, XGBoost, LightGBM, LangChain, Hugging Face Transformers, Pandas, NumPy, Matplotlib, Seaborn, Neo4j, NVIDIA Omniverse
- **Web Development:** HTML5, CSS3, JavaScript, React.js, Vue.js, Flask, FastAPI, Tailwind CSS, REST APIs, WebSockets, JSON, JWT Auth
- **Cloud & Tools:** AWS, Google Cloud, SAS Viya, SAS Studio, Docker, Git/GitHub, CI/CD, Kubernetes, Terraform
- **Programming Languages:** Python, C++, Java, JavaScript, SQL (PostgreSQL, MySQL), R, PHP, Bash/Shell
- **Other Technical Skills:** Computer Vision, Unity Development, Blockchain
- **Soft Skills:** Problem-solving, Team Collaboration, Communication, Agile Methodology, Project Management

EDUCATION

- The University of Sydney (USYD)** - *Master of Computer Science (Data Science & AI)*
Sydney, Australia (2024–2025)
- Indian Institute of Technology Madras** - *B.Sc. in Programming & Data Science (CGPA: 8.02)*
Chennai, India (2020–2023)
- Hansraj College, University of Delhi** - *B.Sc. Physical Science with Computer Science (CGPA: 8.77)*
Delhi, India (2020–2023)

EXPERIENCE

Cloud & Tech Intern

- SAS Institute, Sydney, Jan 2026 – Present
- Supported deployment and troubleshooting of SAS Viya, SAS Studio, and SAS 9.4 across cloud platforms.
 - Improved onboarding flow and resolution times by building internal tools and process guides.
 - Assisted in cross-functional diagnostics and testing for real-time analytics workflows.

Machine Learning Engineer

- AirLabOne, Sydney, Sep 2025 – Dec 2025
- Developed GPU-accelerated ML pipelines using open-source models and NVIDIA Omniverse.
 - Deployed systems for extracting and distributing key insights to user endpoints in production.
 - Tuned performance and reliability in collaboration with software/hardware teams.

Teaching Tutor - Object-Oriented Programming (INFO1113)

- School of Computer Science, USYD, Sydney, Aug 2025 - Present
- Led tutorials/labs on Java and OOP design (classes, inheritance, interfaces, generics) for undergrad.
 - Marked assignments with structured rubrics; delivered targeted feedback during office hours.
 - Authored practice problems and code-review checklists to improve style, testing, and design patterns.

Software Engineering Intern

- ITS Lab, IIT Madras, Chennai, Oct 2023 – Feb 2024
- Built a Python/Flask backend for real-time traffic video analytics, reducing processing latency by ~25%.
 - Shipped a responsive frontend with live metrics (JS), lifting user engagement by ~30%.
 - Implemented secure auth and tuned DB schema for faster queries (~15% improvement).

Backend Developer

Simple Elegant Solutions Pvt. Ltd., New Delhi, Mar 2023 – Apr 2023

- Developed a Python-based data management system, improving handling efficiency (~40%).
- Supported front-end features and collaborated in agile sprints to shorten time-to-market (~15%).

PROJECTS

Graph-QA Agent (Knowledge-Graph Question Answering) — Neo4j, LangGraph, Python, LLMs, Flask

- Engineered a graph-augmented QA pipeline: document ingestion → KG construction → retrieval → reasoning → API.
- Implemented entity/relation extraction and Cypher generation; evaluated grounding and answerability.
- Exposed a minimal Flask Api service with clear interfaces and tests for core components.

Real-Time Traffic Analysis System — Python, OpenCV, AWS

- Detected vehicles in live video feeds (~95% accuracy) and deployed to AWS, lowering latency (~30%).
- Forecasted congestion with classical ML to aid planning.

AI-Powered Heart Disease Prediction — Python (scikit-learn, XGBoost)

- Trained models achieving 90%+ accuracy; added NN baseline; used PCA/SMOTE to address imbalance.

Sports Game Analysis with YOLO (Football/Tennis) — Python (YOLOv5)

- Extracted player/events in real-time; produced stat dashboards for strategy review.

PUBLICATIONS

Antiprotozoal Peptide Prediction

Co-authored a research paper published in Heliyon, focusing on predicting antiprotozoal peptides using various ML algorithms and feature selection techniques.

Result: Achieved a significant reduction in false-positive rates. DOI: [Link](#)

iIL13Pred: Improved Prediction of IL-13 Inducing Peptides

Co-authored a research paper published in BMC Bioinformatics, presenting an advanced prediction model for IL13-inducing peptides.

Result: Achieved a 15% improvement in prediction accuracy compared to existing methods. DOI: [Link](#)

AWARDS & RECOGNITION

- **Winner - SAS Viya for Learners Challenge (ANZ) - 2025**
Top placement in a regional SAS analytics/ML challenge; implemented end-to-end analysis in Viya.
- Winner - Smart India Hackathon (SIH) - 2023, Hansraj College
- Winner - HackLife 2023 (Devpost) - 2023

LEADERSHIP & SERVICE

- Hackathon Mentor - Humanitarian Innovation Hackathon, University of Sydney: mentored teams that placed 1st & 2nd.
- Peer Mentor - ML/DS study support and course strategies.
- Workshop Instructor - "Introduction to JavaScript", Hansraj College.
- Founding Member - Public Policy & Opinion Cell, Hansraj College.
- Group Lead - Kaziranga Group, IIT Madras (collaborative data/SE projects).

ADDITIONAL COURSES & WORKSHOPS

- Machine Learning for Physical Sciences - Hansraj College, University of Delhi (8-week certificate)
- Implementing Machine Learning Techniques - IIT Madras (algorithms & preprocessing)
- Intro to Google Colab & Data Visualisation - IIT Madras